

One Timestamp Does Not Identify Temporal Support in Event-Detection Benchmarks

Supplementary Material

Anonymous CVPR submission

Paper ID *****

This document holds the material referred to from the main paper: the errors-in-variables treatment of the velocity regressor, the protocol behind the occlusion measurement, the data paths of the DSEC measurements, and additional implementation and control analyses. The RVT source revision is b80f5683a6e2d5de65d4bde8105d796ccb50dbb1, and the SSM-ViT source revision is 7c871b55a0c5f00673c2c3975f6b6a1c5adbab88. Section numbers below refer to this document. References labeled main Sec. n refer to the main paper.

1. The velocity regressor

The regression of main Eq. 2 uses the finite-differenced label velocity $\tilde{v} = v + \eta$, where v is latent true velocity and η is label noise. Three error mechanisms enter, and they behave differently. The shared-noise term determines the estimator choice.

Classical dilution. Let the label center carry noise of standard deviation σ_c per frame and let the speed be formed by a central difference over $\pm\Delta$. The differenced speed then carries noise with variance

$$\sigma_\eta^2 = \frac{2\sigma_c^2}{(\Delta_f + \Delta_b)^2}, \quad (1)$$

and τ is attenuated toward zero by $\lambda_{\text{att}} = \text{Var}(v)/(\text{Var}(v) + \sigma_\eta^2)$. The noise is the label noise of the dataset the fit runs on, so it is measured there: Sec. 3 solves for $\sigma_c = 0.6515\text{px}$ on Gen1 itself, and Eq. 1 is evaluated row by row rather than assuming a fixed cadence. On the reported rows $\text{Var}(\eta)$ is 1.0 against a $\text{Var}(\tilde{v})$ of 110.6 in px^2/s^2 , so $\lambda_{\text{att}} = 0.991$ and the correction moves τ by less than 0.2ms. This marginal errors-in-variables calculation indicates small attenuation. The synthetic arm of Sec. 4 injects a known lag under a much larger noise and recovers 0.92 of

it against a predicted 0.933, so the correction is calibrated as well as small. The estimated attenuation is only 0.9%, moving τ by less than 0.2ms, and therefore cannot explain the near-zero coefficient reported here.

Smoothing bias. On the uniform grid used for the label-noise calibration, a central difference returns the mean velocity over 2Δ rather than the instantaneous velocity. Expanding the trajectory about the query instant,

$$\frac{x(t + \Delta) - x(t - \Delta)}{2\Delta} = \dot{x}(t) + \frac{\Delta^2}{6} \ddot{x}(t) + O(\Delta^4), \quad (2)$$

so the regressor carries a systematic term of order $\Delta^2 \ddot{x}/6$. That is a bias in the regressor, not noise in it, and no errors-in-variables correction removes it. It is made small by making Δ small, which Eq. 1 says makes the dilution worse. The reported regression rows use the local secant $[x(t + \Delta_f) - x(t - \Delta_b)]/(\Delta_f + \Delta_b)$. Their median $|\Delta_f - \Delta_b|$ is zero, so the first-order acceleration term is absent in the typical row. The general expansion includes $(\Delta_f - \Delta_b)\ddot{x}/2$ and is covered by the stated local-secant interpretation.

The shared-noise term. The first two mechanisms are properties of the regressor alone. The third couples the regressor to the response. A forward difference $(c_{k+1}^* - c_k^*)/\Delta$ carries the label noise δ_k of the current center with weight $-1/\Delta$, and the residual $\hat{c} - c_k^*$ carries the same δ_k with weight -1 , so

$$\text{Cov}(\hat{c} - c_k^*, v_f) = \frac{\text{Var}(\delta)}{\Delta} \quad (3)$$

even at $\tau = 0$, and the induced slope is $\text{Var}(\delta)/\Delta \text{Var}(v)$. A backward difference carries δ_k with the opposite sign. The centered difference omits c_k^* and removes the direct same-frame shared-noise

term. Residual bias from temporally correlated annotation errors is not identified by this construction, which is why the forward and backward fits are reported beside it as a control.

Instrumental-Variable Alternative. An instrumented estimate would replace the current speed by a velocity finite differenced from a strictly disjoint earlier label pair, on the argument that disjointness implies independent errors. DSEC provides a counterexample to assuming that disjoint label errors are generally independent. The third-difference kernel $(1, -3, 3, -1)$ forces $\rho(1) = -0.75$ for white noise, the same code returns -0.740 on white noise and -0.3566 on the DSEC-Det labels, and pooled-ratio bias, heteroscedastic segments and heavy tails each returned the white-noise value through the identical pooling code, so the departure is a property of the labels. That measurement does not identify what the correlation structure is, beyond excluding a first-order autoregressive form. The paper does not report an instrumental estimate.

The construction term. A benchmark whose high-rate labels are constructed between anchors adds a term to $e_{||}$, where $e_{||} = \langle \hat{c} - c^*, \hat{u} \rangle$, that has nothing to do with the predictor. The chord of a curved image trajectory runs ahead of the truth throughout the interval and meets it at the anchors, which produces a signed, frame-locked, resetting residual for a predictor with no lag. Main Sec. 4.1 shows that exact adjacent-frame linear interpolation is inconsistent with the observed DSEC-Det label trajectories as the sole construction mechanism, and the Gen1 measurements of main Sec. 3 are taken at label times.

2. The influence profile

Main Sec. 3.3 measures the occlusion-sensitivity profile of the released rvt-t checkpoint. The stored input is a stacked histogram of ten bins of 5 ms covering the 50 ms preceding the query time. For each sample the recurrent state is advanced over eight preceding steps so that the network is not read out of a zero state, the reference output is recorded, and then each bin in turn is zeroed and the output recorded again. A bin's influence is the magnitude of the change its occlusion produces in the emitted detection tensor, relative to its norm, averaged over 480 samples drawn from the first twelve lexicographically ordered released Gen1 validation sequences, using the first forty post-warm-up samples from each sequence. This deterministic rule gives 480 samples. The measured occlusion-sensitivity centroid is the weighted average $\sum_k s_k c_k / \sum_k s_k$ over the

ten bin centers c_k , where s_k denotes the measured non-negative bin-ablation sensitivity. The occluded pass starts from the recurrent state entering the step, not from the state the reference pass returned. The two passes therefore share one history and differ only in the occluded bin. If the returned state is used instead, the passes are one step out of phase and the centroid moves by $+1.19$ ms.

For the recurrent-history measurement, the same deterministic sampling rule uses 576 samples, the first forty-eight post-warm-up samples from each of the same twelve validation sequences. The current window and the preceding nine windows form the 500 ms horizon, and the current window and preceding nineteen windows form the 1000 ms horizon. For each lag, the corresponding whole input window is zeroed while the saved reference history before that lag is restored, and the state is propagated through the intervening windows to the target step. The relative L_2 change in the target detection tensor is averaged over samples to give a non-negative window sensitivity q_ℓ . The horizon centroid is

$$c_{q,H} = \frac{\sum_{\ell=0}^{L_H-1} q_\ell [-(25 + 50\ell) \text{ ms}]}{\sum_{\ell=0}^{L_H-1} q_\ell},$$

where L_H is the number of windows in horizon H , and the newest-window share is $q_0 / \sum_\ell q_\ell$. The same saved reference history is used for the reference and ablated passes at each lag.

Two properties of that measurement bound it. Occlusion reports the sensitivity of the output to a bin. The ablation profile is an operational input-sensitivity measure and is not, in general, a decomposition of temporal weights. For a nonlinear predictor it is a local rather than a global description. And a single-bin occlusion leaves the remaining bins in place, so the measurement is a partial derivative rather than a decomposition of the output.

Masking the leading bins of the stored input is not a route to the same quantity at a shorter support. For a fixed network of influence weights a_k on bin centers c_k , let a constant-velocity trajectory be $p(t) = p_0 + vt$. The conditional expectation after masking the leading j bins is $A_j p_0 + C_j v$ with $A_j = \sum_S a_k$ and $C_j = \sum_S a_k c_k$ over the retained set S . Reading a time off that requires the normalization C_j / A_j , which the network need not perform, and the result equals the retained-mass centroid only for non-negative weights.

3. The label-center noise of Gen1

The errors-in-variables treatment of Sec. 1 and the calibration of the velocity estimator both need the dis-

persion of the Gen1 label center about a smooth trajectory. The value solved for elsewhere in this work, 0.4974px, is DSEC-Det's, and does not transfer to a different annotation effort on a different sensor, so it is solved again on Gen1.

For a track center series $x_i = f(t_i) + e_i$ on a uniform grid, the k -th difference has $\text{Var}(D_k) = \binom{2k}{k} \sigma^2$ plus a motion term of order k , so $\text{Var}(D_k)/\binom{2k}{k}$ falls toward σ^2 from above. Halving the sampling rate multiplies the motion term by 4^k and leaves σ^2 untouched, giving a second equation in the same two unknowns, $\sigma^2 = (4^k V_{\text{full}} - V_{\text{half}})/(4^k - 1)$. Tracks are rebuilt by the same-class overlap linking the output-time measurement uses, giving 691 tracks of at least twelve labels on uniform annotation grids with cadences of 250, 500, or 1000 ms, corresponding to 4, 2, or 1 Hz. The reported regression rows instead use the two adjacent label neighbors with variable intervals. Their row-wise noise variance is $2\sigma_c^2/(\Delta_f + \Delta_b)^2$, rather than a fixed-cadence approximation. Variances are trimmed at 5%, because all four Gen1 box coordinates are stored as integers and a median absolute deviation on lattice-valued data locks onto discrete values.

The two-rate elimination agrees with the full-rate ladder to four decimals from $k = 5$ onward, which is the check that the motion term is negligible rather than assumed away. The ladder extrapolates to $\sigma = 0.6515$ px, of which 0.1443 px is the quantization of a 0.5 px center lattice, leaving 0.6353 px of annotation dispersion.

4. The sign-flip calibration

Equation 3 predicts an artifact whose size is set by the label noise and the observed velocity variance, so it is calibrated before the estimator is used. 2000 synthetic tracks of fourteen labels on the Gen1 grid are generated with quadratic motion. The synthetic experiment calibrates the shared-noise artifact and attenuation under quadratic motion. The simulated detector reports the true position at $t - \tau_{\text{true}}$ with its own independent error, and the label centers are corrupted at 0.6515 px.

With $\tau_{\text{true}} = 0$ the forward estimator returns -5.64 ms and the backward one $+5.84$ ms, summing to $+0.20$ ms, against the closed form -5.63 ms, the centered estimator returns $+0.00$ ms. Setting the label noise to zero returns all three to $+0.02$ ms, and the placebo stays at its null throughout, so it does not absorb the artifact and remains usable on real rows. In the synthetic physical-lag experiment, the centered estimator recovers 0.92 of the injected lag, consistent with the predicted attenuation of 0.933 for this noise.

5. The output-time specification sweeps

Lag Recovery under the Full Control Set. Adding $-\tau_{\text{inj}} \tilde{v}$ to the observed residual creates a known perturbation along the fitted temporal regressor. The coefficient is recovered at 1.000 of its injected value at 10, 25, 50 and 100 ms, showing that the geometric controls do not absorb this regressor. Injecting instead a purely geometric displacement of 0.05 px per px of box height moves the minimal specification's τ by $+0.63$ ms and the reported one's by less than 0.01 ms.

Sensitivity to Axis Pairing. The reported specification pairs box height and image row with $\hat{y}$ and box width and image column with $\hat{x}$. Entering all four on both axes gives -3.62 ± 7.41 ms. Dropping each in turn moves τ by under 0.3 ms except for the image column, whose removal returns the placebo to its inflated value. Raising the overlap floor to 0.7 gives $+0.49 \pm 4.67$ ms, the tightest of the sweep. Across every specification carrying geometric controls, including speed floors from 5 to 20 px/s, τ spans -20.2 to $+11.1$ ms, and none of them reaches the occlusion-sensitivity centroid.

The bootstrap. Resampling sequences with replacement, 4000 times, gives a 95% interval of $[-17.0, +11.5]$ ms with 3 draws reaching the influence centroid. The fit absorbs the per-sequence constant vectors by within-cell demeaning, which reproduces the full design's coefficient to 0.007 ms.

6. The per-object ratio

For comparison with the estimator in main Eq. 2, we evaluate the per-object ratio $-(\hat{c} - c^*, \hat{u})/\|\tilde{v}\|$ on the same 22534 rows. Across these rows, it has a sample standard deviation of 4170 ms and an interquartile range of 430 ms. Its mean over all rows is $+24.22 \pm 27.73$ ms, close to the $+23.81$ ms coefficient corresponding to the newest-window occlusion-sensitivity centroid under the convention of main Eq. 2. Above a 15 px/s floor it gives $+13.34 \pm 6.17$ ms, significant at more than two standard errors, which is a fixed spatial bias divided by a speed: the ratio has no intercept and main Eq. 2 does.

7. The port of the released checkpoint

The released rvt-t checkpoint loads into a model built from the repository's own configuration files with 0 missing and 0 unexpected keys at 4.41 M parameters, and runs forward under a current framework without the training code it was serialized from. Both predictor measurements run on that port.

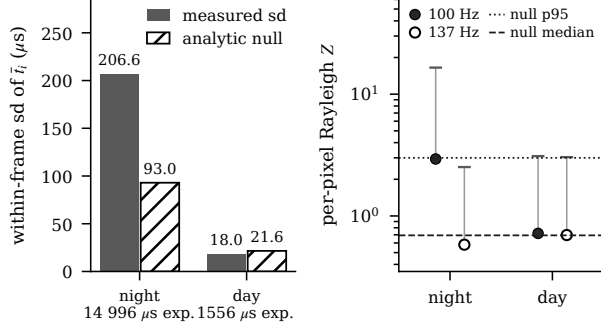


Figure 1. Left: within-frame dispersion of $\bar{t}_i$ against its analytic null in the two exposure strata, 1134 frames of `zurich_city_09_a` at 14996 μ s and 20 frames of `interlaken_00_c` at 1556 μ s. The daytime frames are selected by retaining frames with at least three qualifying boxes. Right: per-pixel Rayleigh statistic $Z = R^2/N$ at 100.00 Hz and at 137 Hz for the same two sequences, median with a whisker to the ninety-fifth percentile, against the Exp(1) null. Both contrasts run in the same direction, and Main Sec. 4.4 reports the controls applied to that dispersion.

Arm	frames	sd	null	excess
Ceiling sequence, all pixels	1134	206.6	93.0	183.6
Phase-locked pixels removed	1134	205.1	97.1	180.6
Random pixels, matched size	1133	206.3	94.8	182.9
Random-position boxes	1123	338.2	134.5	306.3
by each box's own 100.00 Hz modulation depth m				
$m \in [0.122, 0.410)$	270	119.3	74.6	93.1
$m \in [0.410, 0.467)$	243	149.1	76.2	128.1
$m \in [0.467, 0.534)$	207	156.6	85.1	131.5
$m \in [0.534, 1.050]$	265	227.2	91.5	208.0

Table 1. Per-object event centroid inside one published exposure window of `zurich_city_09_a`, in μ s, each entry the median over frames of a within-frame statistic against that frame's own null $w^2/12N_i$. The first block's excess follows the main Sec. 4.4 statistic summarized over frames. Rows two and three remove the 14334 pixels phase-locked at 100.00 Hz and a random subset of that size, row four places boxes of the same shapes at random positions. In the lower block a frame enters a stratum when 3 of its boxes fall in it, so those frame counts do not partition the first row's.

8. The DSEC measurements

Data path. For each labeled frame of a sequence, the frame's own exposure interval $[a, b]$ is read from the dataset's published `image_exposure_timestamps_left` file, the event stream is sliced to that interval through the release's `ms_to_idx` index, and the events inside each released box are collected. The `ms_to_idx` index resolves to whole milliseconds, so the slice it returns is filtered to $a \leq t < b$ in microseconds before any statistic is formed, and the analytic null is computed for the same interval. Events and frames share one clock in this dataset, so no calibration step enters. For the dispersion of main Sec. 4.4 a box contributes only if it

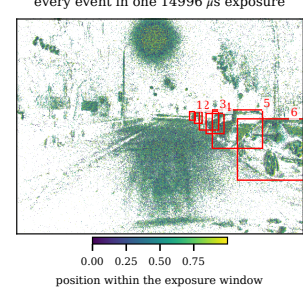


Figure 2. Every event inside one published 14996 μ s exposure window of `zurich_city_09_a`, colored by its position within that window, with that frame's released boxes numbered left to right. The frame is the median-dispersion one of 602 eligible candidate frames with at least six qualifying boxes, not the largest.

Sequence	boxes	m	$Z_{100.00}$	Z_{137}	locked
<code>zurich_city_00_a</code>	2914	0.428	117.6	3.15	100.00 %
<code>zurich_city_01_a</code>	2134	0.440	99.1	4.69	99.63 %
<code>zurich_city_02_a</code>	619	0.445	68.4	2.75	99.68 %
<code>zurich_city_03_a</code>	241	0.489	310.9	25.85	95.44 %
<code>zurich_city_09_a</code>	6997	0.466	251.5	7.82	99.23 %
<code>zurich_city_10_a</code>	3611	0.438	111.6	3.33	100.00 %

Table 2. Every DSEC training sequence whose exposure is pinned at 14996 μ s, measured per labeled box over the events inside that box during that frame's own published exposure window, with at least 400 events per box. m is the median over boxes of the modulation depth $m = 2|C|$ at 100.00 Hz. $Z_{100.00}$ and Z_{137} are the median over boxes of the Rayleigh statistic at 100.00 Hz and at the off-frequency control on the same events. The analytic Exp(1) median of 0.693 does not describe either column, which is why the two are compared with each other. The locked fraction denotes the fraction of boxes whose local excess at 100.00 Hz exceeds the ninety-ninth percentile of the same quantity at 137 Hz.

carries at least 200 events in the window, and a frame only if at least 3 boxes qualify, the reported statistics are medians over frames of a within-frame quantity, so no frame is weighted by its object count. For the per-box statistics of Table 2 a box contributes with at least 400 events and no frame-level requirement applies. The selected qualitative frame is shown in Supplement Fig. 2, and the paired daytime control is shown in Supplement Fig. 1.

For frame f , let I_f be the qualifying boxes, let N_{fi} be the event count inside box i , and let $\bar{t}_{fi} = N_{fi}^{-1} \sum_{k \in i} t_k$ be its event-time centroid relative to the frame's mid-exposure. The observed within-frame dis-

persion and its frame-specific analytic null are

$$d_f = \left[\frac{1}{|I_f| - 1} \sum_{i \in I_f} (\bar{t}_{fi} - \bar{t}_f)^2 \right]^{1/2},$$

$$n_f = \left[\frac{1}{|I_f|} \sum_{i \in I_f} \frac{w_f^2}{12N_{fi}} \right]^{1/2}.$$

where $\bar{t}_f = |I_f|^{-1} \sum_{i \in I_f} \bar{t}_{fi}$ and $w_f = b_f - a_f$. The frame-level excess is $e_f = [\max(d_f^2 - n_f^2, 0)]^{1/2}$, and the reported entries are the medians of d_f , n_f , and e_f over eligible frames.

The stereo exposures. The left and right exposures open at the same microsecond in 100.00 % of the 7080 frames checked and differ only in when they close, and the published image timestamp equals the mean of the two mid-exposures to within 1 μ s.

Archive version and track association. DSEC-Det ships more than one annotation version. The numbers in main Sec. 4.1 describe the archive dsec-det_left_object_detections.zip, 4655023 bytes, retrieved 2026-09-01. Whether those labels are tracker output on the RGB frames, which the dataset's own description states, is not distinguishable from human annotation by a second-difference residual. On the Gen1 side the track_id field is a per-frame index rather than a persistent identifier, so association for the output-time fit is rebuilt by same-class overlap between adjacent label times, at which 79.0 % of boxes link.

Null. For events uniform on $[a, b]$ the mean of N_i times has variance $w^2/12N_i$ with $w = b - a$. Each frame therefore carries its own null, formed from the published width and the observed counts, and the excess of main Sec. 4.4 is summarized over frames rather than formed from the medians of the two summaries. That is why the excess column of Table 1 is not $(\text{sd}^2 - \text{null}^2)^{1/2}$ of the two columns beside it.

Phase test. The per-pixel test at frequency f takes the event phases $\phi_k = 2\pi f t_k \bmod 2\pi$ of a pixel, forms the resultant length R and reports $Z = R^2/N$, which is $\text{Exp}(1)$ under uniform phase. Raw modulation depth is not usable at these counts, because for N events at random phase the resultant is already of the size of the effect. Pixels enter with at least 100 events over the tested slice, and every fourth event is used to reduce sample size. The off-frequency and daytime controls remain close to their nominal reference under the

same procedure. Two arms hold the estimator to its null, an off-frequency arm on the same events and the same frequency on daytime events. The fraction of locked pixels is reported relative to live pixels in main Sec. 4.2 and separately relative to the full sensor. The two denominators differ, and each statement identifies its denominator.

The per-box statistics of Table 2. The same estimator is applied to the events inside one released box during one published exposure window rather than to the events of one pixel over a long slice. For box i the depth is $m = 2|C|$ and the statistic is $Z = N_i|C|^2$ with $\omega = 2\pi f$ and $C = N_i^{-1} \sum_{k \in i} e^{-i\omega t_k}$, phases referred to that frame's mid-exposure instant. A finite exposure does not produce uniform phase at a tested frequency, so the pixel-level $\text{Exp}(1)$ null is not assumed for these per-box values. The depth is reported descriptively, without a random-phase floor. Boxes are matched to exposure windows by nearest mid-exposure instant within half a frame period. The off-frequency arm at 137 Hz is run on exactly the same events.

Pixel exclusion and modulation strata. The exclusion mask of Table 1 is the set of pixels reaching $p < 10^{-3}$ in the per-pixel test, and the matched control draws a random pixel set of exactly the same size, so the two arms remove the same number of pixels. The mask is built from a six-second slice at every fourth event. The stratification uses no mask: the quartiles of m are taken over all 6841 qualifying boxes, and a frame contributes to a stratum when at least 3 of its boxes fall inside it, which is why the frame counts of that block are not a partition of the first row's.

9. The permutation null

The analytic null $w^2/12N_i$ for the main Sec. 4.4 statistic is the variance of a mean of N_i draws uniform on the exposure, so it assumes the events of a box arrive independently and uniformly.

The permutation null does not assume independent or uniform event arrivals. Instead, it conditions on the pooled frame-level event times and each box's event count while randomizing box assignment. For each frame the events of all qualifying boxes are pooled and reassigned to boxes at random, preserving each box's count. The exposure, the flicker and the scene's burstiness all survive that reassignment, only which box an event belongs to is destroyed. Over 1134 frames and 200 draws each it returns 78.6 μ s, 0.85 times the analytic null rather than above it, and 94.1 % of frames exceed their own permutation null. A burst common to

the frame displaces every $\bar{t}_i$ in the same direction and cancels in their dispersion, which is why the statistic was already insensitive to it. For a separate random-position draw evaluated against the permutation null, boxes of the same shapes give $339.4\mu\text{s}$ against $113.3\mu\text{s}$. Thus the dispersion is not specific to annotated object regions, although object regions may still affect it.

10. The carried state of the SSM release

Main Sec. 3.3 measures five released checkpoints of two architectures. The second is SSM-ViT (Zubic et al., CVPR 2024), a fork of the RVT repository in which each stage’s ConvLSTM is replaced by an S5 layer. It distributes RVT’s own preprocessed Gen1 and builds its window boundaries with the same statement of the same script, so the window convention of main Sec. 3.2 holds there unchanged.

Its recurrent state does not. Let s denote the carried state entering the first step of a chunk. That release injects it as $\bar{\Lambda}_0 \leftarrow \bar{\Lambda}_0 \odot s$ before an associative scan whose binary operator is $(a_i, b_i) \circ (a_j, b_j) = (a_j a_i, a_j b_i + b_j)$. The scan’s outputs are the b components, and b_p is built from $a_1, \dots, a_p$ and $b_0, \dots, b_p$, $\bar{\Lambda}_0$ enters only the a component of the first element and therefore no output. The carried state is inert.

We verify this code-level property empirically. Driving each model the way its own release drives it and zeroing the state entering a chunk of 8 windows, the relative L_2 change in the emitted detection tensor is 0.00000 at every position for SSM-ViT, and 0.0616 at position one falling to 0.0304 at the last position for rvt-t. A model that ignored its history altogether would return zero to both arms, so the control occludes the window one position earlier inside the same chunk: SSM-ViT answers 0.0276 there, falling to 0.0106, so it is recurrent within a chunk and only across chunks is the state lost.

11. Effect of Temporal Displacement on Checkpoint Ordering

Main Sec. 3.5 reports that displacing the ground truth to the newest window’s occlusion-sensitivity centroid costs 0.04 points on one checkpoint. Whether the interval changes a comparison needs more than one detector, so all 5 released checkpoints of main Sec. 3.3 were run over the Gen1 validation split under one protocol: the same frames, the same confidence threshold and non-maximum suppression, and ground truth that is identical across the five and identical to the dump main Sec. 3.5 already uses. Each was then scored at the label instant and again at the displacement it prefers, which ranges from -15 to 0 ms across the five.

stratum	ordering at the label instant
all moving	$b > s > t > S5-B > S5-S$
10–25 px/s	$S5-B > b > t > S5-S > s$
25–50 px/s	$S5-B > b > S5-S > s > t$
≥ 50 px/s	$b > t > S5-S > s > S5-B$

Table 3. Ordering of the 5 released checkpoints at the label instant, by object speed, t , s , b are the RVT capacities and S5-S, S5-B the SSM ones. Scoring each at its own preferred displacement leaves every one of these orderings unchanged. A common displacement sweep reveals one localized inversion between rvt-b and S5-S in the 25–50 px/s stratum. Ranking varies across speed strata, and the common sweep produces no other pairwise inversion. s5vit-base is first in the middle two strata and last in the fastest, and rvt-t is third overall, last between 25 and 50 px/s and second above 50.

The own-preferred-offset scores preserve the reported ordering in each speed stratum. Under a common displacement sweep, the 25–50 px/s stratum contains a localized ranking inversion between rvt-b and S5-S. The other evaluated strata contain no sign-changing pair. The aggregate ranking and the fast-object ranking have different leaders, so the strata are reported separately.

The released streaming evaluation steps through the split in non-overlapping chunks of 21 windows (sequence_length in config/dataset/gen1.yaml, start_indices in sequence_for_streaming.py steps by exactly that). The evaluated input sensitivity of a released detection therefore depends on where its frame falls inside that chunk, from a full 21 windows at the last position to the current window alone at the first. This is reported as a property of the release that its documentation does not state, which is the same class of undeclared quantity the rest of this paper measures, and not as a claim about what the architecture could support if the state were carried.